

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

December 2011

CL 302 Structural Analysis-II

Date: 13.12.2011, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 Analyze the continuous beam shown in **Figure 1** and draw shear force and bending moment diagrams. [07]
- Q - 2 (a) Sketch ILD for M_x @ $x=2\text{m}$ giving ordinates at 1m intervals for the two span continuous beam shown in the **Figure 2**. Assume EI is constant throughout. [09]
- (b) Analyze the Fixed beam shown in **Figure 3** and draw shear force and bending moment diagrams. [05]

OR

- Q - 2 (a) Sketch ILD for FX @ $x=1\text{m}$ giving ordinates at 1m intervals for the propped cantilever shown in the **Figure 4**. [09]
- (b) Analyze the Fixed beam shown in **Figure 5** and draw shear force and bending moment diagrams. [05]
- Q - 3 (a) State Muller-Breslau Principle. [02]
- (b) Attempt any two [12]
1. A cable is used to support five equal and equidistant loads over a span of 30 m. Find the length of the cable required and its sectional area if the safe tensile stress is 140 N/mm^2 . The central dip of the cable is 2.5 m and loads are 5 kN each. Also find tensions in each part.
 2. A cable loaded with 10 kN/m is stretched between two supports in the same horizontal line 200m apart. If the central dip is 15m, find the maximum and minimum pulls in the cables.
 3. A light cable 18 m long is supported at two ends at the same level. The supports are 16 m apart. The cable supports three loads 8, 10 and 12 kN dividing the 16 m distance in four equal parts. Find shape of the cable and tension in various parts.

SECTION - II

- Q - 4 A compound cylinder is made by shrinkage a tube of 160 mm internal diameter and 20 mm thick over another tube of 160 mm external diameter and 20 mm thick. The radial pressure at the common surface, after shrinkage, is 8 N/mm^2 . Find the final stresses set up across the section, when the compound cylinder is subjected to an internal fluid pressure of 60 N/mm^2 . [07]
- Q - 5 (a) A thick metallic cylindrical shell of 150 mm internal diameter is required to withstand an internal pressure of 8 kN/mm^2 . Find the necessary thickness of the shell, if the [03]

permissible tensile stress in the section is 20 N/mm^2 .

- (b) Derive an equation for the strain energy stored in a body due to impact loading. [03]
- (c) A vertical round steel rod 1.82 meter long is securely held at its upper end. A weight can slide freely on the rod and its fall is arrested by a stop provided at the lower end of the rod. When the weight falls from a height of 30 mm above the stop the maximum stress reached in the rod is estimated to be 157 N/mm^2 . Determine the stress in the rod if the load had been applied gradually and also the minimum stress if the load had fallen from a height of 47.5 mm. Take $E = 2 \times 10^5 \text{ N/mm}^2$. [04]
- (d) Determine reaction at B for a propped cantilever beam using Castigliano's second theorem as shown in Figure 6. [04]

OR

- Q - 5 (a) In a thick walled cylindrical shell, the radial pressure is 20 MPa at a radius of 400 mm and 5 MPa when the radius is 600 mm. If the internal radius of the cylinder is 200 mm, determine the maximum radial pressure and maximum hoop stress. [03]
- (b) Write the strain energy expressions for following conditions: [03]
- The load is applied with impact.
 - The load is applied with bending
 - The load is applied with shear
- (c) Determine the strain energy of a cantilever beam of span 2 m having size 20 mm width x 60 mm depth. Take $E = 200 \text{ GPa}$. [04]
- When 1000 N concentrated load is placed at free end.
 - When total 1000 N load is uniformly distributed over the entire span.
- (d) Calculate horizontal displacements of point C for a rigid cantilever frame shown in Figure 7. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 3 \times 10^8 \text{ mm}^4$. [04]

Q - 6 (a) Attempt any one: [04]

- Find deflection at B for a cantilever beam loaded as shown in Figure 8. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 3 \times 10^8 \text{ mm}^4$.
 - Find slope at D for a cantilever beam loaded as shown in Figure 8. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 3 \times 10^8 \text{ mm}^4$.
- (b) Solve any two beams by method of consistent deformation from Figure 9, Figure 10 & Figure 11. [10]

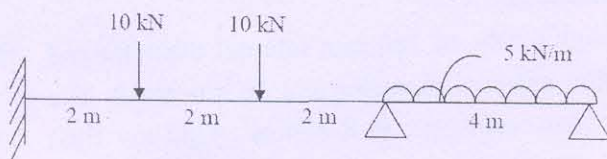


Figure 1



Figure 2

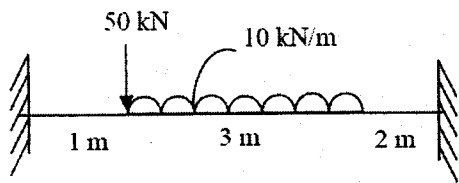


Figure 3

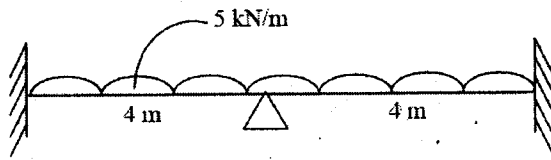


Figure 5

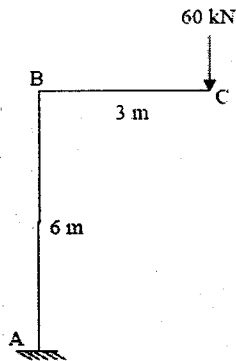


Figure 7

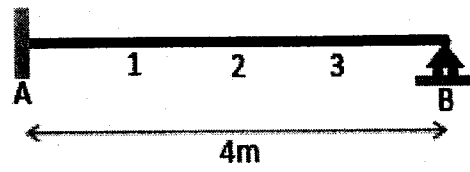


Figure 4

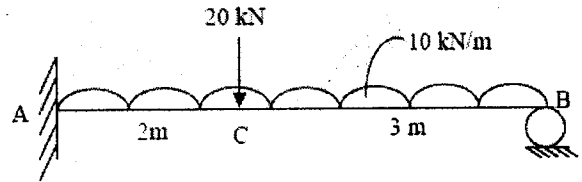


Figure 6

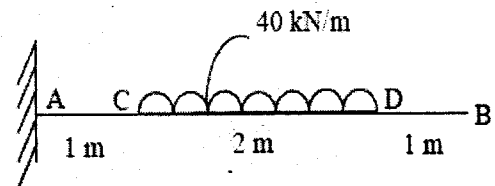


Figure 8

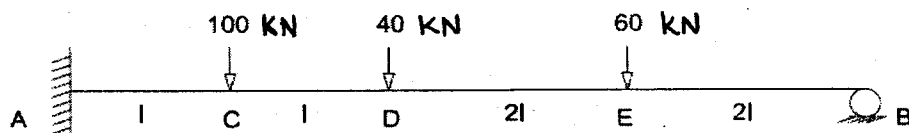


Figure 9

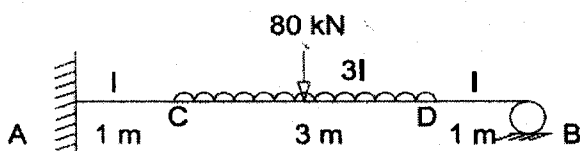


Figure 10

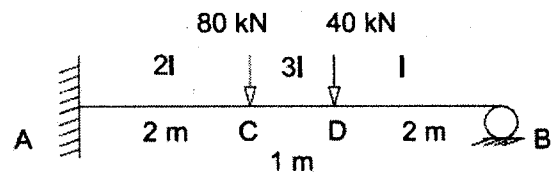


Figure 11
